Hyojoon Park

PH.D. CANDIDATE | 3D GRAPHICS/VISION, ML/COMPRESSION/SIMULATION

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Professional Summary

- My research integrates **Machine Learning** with **Computer Graphics** and **Computer Vision**, building end-to-end systems that combine physics-based simulation, geometry, and image/video processing.
- My work spans **3D reconstruction** of bodies, faces, and scenes, from camera calibration and bundle adjustment to **3D Gaussian Splatting (3DGS)**, along with **neural simulation** of face and hair, hand reconstruction for **XR (VR/AR)**, and **haptics** for dental simulation.
- I develop **4D medical image and video compression with deep learning** using spatiotemporal wavelets, hyperprior entropy models, latent quantization, and rate-distortion optimization.

Education

University of Wisconsin-Madison — *Madison, WI, USA*

Ph.D. Candidate in Computer Sciences — *Sep. 2021 - Expected Dec. 2026*

- **Research:** Machine learning for physics-based simulation, computer graphics, and 4D medical image/video compression
- **Advisors:** Eftychios Sifakis and Kevin Eliceiri

Seoul National University — *Seoul, Korea*

M.S. in Mechanical Engineering — *Mar. 2017 - Feb. 2019*

- **Thesis:** Dental Simulator with Increased Z-width of Haptic Rendering (also presented at AsiaHaptics 2018)
- **Advisor:** Dongjun Lee
- **Outstanding MS Thesis Presentation Award** [M.S. thesis presentation]

Technical Contributions

- Extended **passive decomposition** to SE(3) using **passive midpoint integration** for stable, stiff haptic rendering.
- Derived a **momentum-based human force observer** to estimate human forces on multi-DoF haptic devices without force sensors.
- Implemented **octree voxel rendering** for real-time collision detection and material removal in virtual tooth drilling.

Korea University — *Seoul, Korea*

B.S. in Mechanical Engineering — *Mar. 2010 - Feb. 2017*

- Apr. 2014 - Jul. 2014: **Technical University of Munich (TUM)** - Exchange Student
- Jun. 2011 - Mar. 2013: **Republic of Korea Army (ROKA)** - Military Service

Work Experience

Google — *Playa Vista, CA, USA*

Student Researcher — *Sep. 2025 - May 2026*

- Developed **real-time hair animation** for **digital humans** and XR with a **self-supervised, physics-based neural hair simulator**. Publication-track results in progress.

Technical Contributions

- Designed a **GRU motion encoder** to model hair inertia, natural swinging, and settling from motion history.
- Encoded local and global hair-groom structure with a **latent transformer** that scales linearly with strand count.
- Built a **StyleGAN2**-inspired **modulated MLP** decoder conditioned on body shape, hairstyle, and motion history for hair animation.

Nokia Bell Labs — *New Providence, NJ, USA*

Industrial Metaverse Intern — *Jun. 2025 - Aug. 2025*

- Built a real-time **3DGS digital twin** with **language-guided 3D interaction** and physical registration, placing **first among 100+ global interns**. Follow-up research studies capture-geometry effects on reconstruction quality. Publication-track results in progress.

Technical Contributions

- Registered 3DGS to real-world coordinates using ChArUco triangulation, reprojection optimization, and Umeyama alignment.
- Embedded language features in 3D Gaussians and used DBSCAN for real-time, **text-guided 3D object segmentation**.

NVIDIA — *Santa Clara, CA, USA*

Software Intern - Physics-Based Simulations — *May 2024 - Aug. 2024*

- Developed **single-image 3D face reconstruction** for **identity-preserving digital humans**, re-aging faces from infancy to old age.

Technical Contributions

- Generated age-specific training meshes through **deformation transfer** from scanned faces to age-specific templates.
- Learned the **3D morphable model (3DMM)** identity basis from skull-aligned face meshes centered on age-specific reference shapes.
- Trained an end-to-end network to predict 3DMM identity and age parameters from a single image using a pretrained MiVOLO age prior, **differentiable rendering**, and **GAN-based training**.

University of Utah
Salt Lake City, UT, USA

Graduate Research Assistant
Sep. 2019 - Jun. 2021

- Developed and open-sourced a multi-view body-reconstruction system for SIGGRAPH 2021, recovering 1,000+ uniquely labeled points from RGB cameras and a low-cost patterned suit to capture fine skin deformation. [code] [paper]

  Technical Contributions
  - Built a **multi-camera calibration and bundle adjustment** pipeline in C++ and Python with Ceres Solver.
  - Built a **VAE outlier detector** to filter erroneous checkerboard corners and improve calibration.
  - Trained **CNN regressors and classifiers** for corner localization, candidate validation, and code recognition.

Korea Institute of Science and Technology (KIST)
Seoul, Korea

Research Intern
Mar. 2019 - Aug. 2019

- Developed an IROS 2020 VR hand-rendering system that reconstructs realistic full-hand poses from fingertip positions. [paper] [video]

  Technical Contributions
  - Calibrated virtual hand size from fingertip measurements and tracking-device geometry using SVD-based similarity alignment.
  - Applied **motion retargeting** with per-joint scaling and offsets calibrated from open-hand and grasping poses.
  - Extended **FABRIK inverse kinematics** with planar hinge constraints for natural finger bending during **real-time hand reconstruction**.

Publications

Near-realtime Facial Animation by Deep 3D Simulation Super-Resolution
***Hyojoon Park**, Sangeetha Grama Srinivasan, Matthew Cong, Doyub Kim, Byungsoo Kim, Jonathan Swartz, Ken Museth, Eftychios Sifakis*
ACM Transactions on Graphics (TOG), 2024 (Presented at SIGGRAPH ASIA 2024) [paper] [code] [featured video]

- Combines coarse simulation with **neural 3D super-resolution**, producing detailed facial animation at 18.46 FPS (115x faster than the traditional simulator).
- Uses **graph neural networks** and **coordinate-based MLPs** for volumetric encoding and geodesic interpolation of coarse simulation output, generalizing from quasi-static training data to unseen expressions, dynamics, and external forces.

Collagen Fiber Centerline Tracking in Fibrotic Tissue via Deep Neural Networks with Variational Autoencoder-based Synthetic Training Data Generation
Hyojoon Park*, *Bin Li*, Yuming Liu, Michael S. Nelson, Helen M. Wilson, Eftychios Sifakis, and Kevin W. Eliceiri (*equal contributions)*
Medical Image Analysis, 2023 [paper] [code]

- Introduces **DuoVAE**, pairing two independent **VAEs** for **controllable synthetic data generation** across diverse collagen-fiber morphologies.
- Combines a **conditional GAN** for image synthesis with **U-Net** centerline extraction for collagen-fiber centerline tracking.

Capturing Detailed Deformations of Moving Human Bodies
*He Chen, **Hyojoon Park**, Kutay Macit, and Ladislav Kavan*
SIGGRAPH, 2021 [paper] [code]

- Reconstructs 1,000+ uniquely labeled body points from 16 RGB cameras using **CNN regressors and classifiers** for corner localization, candidate validation, and code recognition, followed by **multi-view triangulation**, without temporal tracking for point identification.
- Captures breathing, muscle motion, and flesh deformation with pixel-level reprojection accuracy through camera calibration and **bundle adjustment**.

Adaptive Precision-Enhancing Hand Rendering for Wearable Fingertip Tracking Devices
***Hyojoon Park** and Jung-Min Park*
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020 [paper] [video]

- Extends **FABRIK inverse kinematics** with planar hinge constraints for real-time full-hand reconstruction from fingertip positions and uses **motion retargeting** to improve pose alignment.
- Reduces grasping-task time by 17.08% for large objects and 28.14% for small objects versus rendering the hand as spheres.

Dental Simulator with Increased Z-width of Haptic Rendering
***Hyojoon Park**, Myungsin Kim, and Dongjun Lee*
AsiaHaptics, 2018 [paper] [video]

- Uses **passive midpoint integration** to substantially increase stable virtual-coupling stiffness for high-fidelity haptic rendering.
- Extends **passive decomposition** to 6-DoF haptics with a momentum-based human force observer, reducing device-proxy coordination error from 20 mm to near zero.

Publications (continued)

Rigid-body Collaborative Manipulation among Remote Users with Wearable Cutaneous Haptic Interfaces
*Myungsin Kim, WonHa Lee, **Hyojoon Park**, Junghan Kwon, Yong-Lae Park, and Dongjun Lee*
AsiaHaptics, 2018 [paper] [video]
- Enables two remote users to manipulate a shared rigid body in VR with wearable cutaneous feedback.

Stretchable Skin-Like Cooling/Heating Device for Reconstruction of Artificial Thermal Sensation in Virtual Reality
*Jinwoo Lee, Heayoun Sul, Wonha Lee, Kyung Rok Pyun, Inho Ha, Dongkwan Kim, **Hyojoon Park**, Hyeonjin Eom, Yeosang Yoon, Jinwook Jung, Dongjun Lee, and Seung Hwan Ko*
Advanced Functional Materials, 2020 [paper]
- Reproduces hot, cold, and material-dependent thermal cues at three fingertips in VR using stretchable heating/cooling devices.

Highly Stretchable and Oxidation-resistive Cu Nanowire Heater for Replication of the Feeling of Heat in a Virtual World
*Dongkwan Kim, Junhyuk Bang, Wonha Lee, Inho Ha, Jinwoo Lee, Hyeonjin Eom, Myungsin Kim, Jungjae Park, Joonhwa Choi, Jinhyung Kwon, Seungyong Han, **Hyojoon Park**, Dongjun Lee, and Seung Hwan Ko*
Journal of Materials Chemistry A, 2020 [paper]
- Reproduces contact and radiative heat in VR using a 12-pixel thermal glove with motion tracking and closed-loop temperature control.

Wearable Cutaneous Haptic Interface with Soft Sensors and IMUs
*Yongjun Lee, Myungsin Kim, Yongseok Lee, **Hyojoon Park**, and Dongjun Lee*
Korea Robotics Society Annual Conference, 2018

Design and Performance Evaluation of Wearable Haptic Interfaces
*WonHa Lee, Myungsin Kim, **Hyojoon Park**, and Dongjun Lee*
International Conference on Control, Automation and Systems, 2018

Professional Service

SIGGRAPH Asia 2026 Reviewer *2026*

Technical Skills

Languages	Python, C++, C, CUDA
Frameworks	PyTorch, JAX, OpenCV, NumPy/SciPy, OpenGL, Ceres Solver
Core Areas	Machine Learning, Deep Learning, Computer Vision, Computer Graphics, 3D Vision, 3D Reconstruction, Neural Rendering, Physics-based Simulation, Medical Imaging, Image/Video Compression, XR, Haptics
Methods	Transformers, GNNs, CNNs, GRUs, MLPs, GANs, VAEs, StyleGAN2, U-Net, 3DGS, Differentiable Rendering
Tooling	Linux, Git, Docker, CMake, LaTeX

Teaching Experience

Computer Graphics (CS559) *University of Wisconsin-Madison* *Spring 2022*
Computer Graphics (CS559) *University of Wisconsin-Madison* *Fall 2021*
Interactive Computer Graphics (CS6610) *University of Utah* *Spring 2021*
Computer Graphics (CS4600) *University of Utah* *Fall 2020*
System Analysis in Mechanical Engineering *Seoul National University* *Spring 2018*